

'Black Box' to 'Conversational' Machine Learning: Ondansetron Reduces Risk of Hospital-Acquired Venous Thromboembolism

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Abstract—Machine learning, combined with a proliferation of electronic healthcare records (EHR), has the potential to transform medicine by identifying previously unknown interventions that reduce the risk of adverse outcomes. To realize this potential, machine learning must leave the conceptual 'black box' in complex domains to overcome several pitfalls, like the presence of confounding variables. These variables predict outcomes but are not causal, often yielding uninformative models. In this work, we envision a 'conversational' approach to design machine learning models, which couple modeling decisions to domain expertise. We demonstrate this approach via a retrospective cohort study to identify factors which affect the risk of hospital-acquired venous thromboembolism (HA-VTE). Using logistic regression for modeling, we have identified drugs that reduce the risk of HA-VTE. Our analysis reveals that ondansetron, an anti-nausea and anti-emetic medication, commonly used in treating side-effects of chemotherapy and post-general anesthesia period, substantially reduces the risk of HA-VTE when compared to aspirin (11% vs. 15% relative risk reduction or RRR, respectively). The low cost and low morbidity of ondansetron may justify further inquiry into its use as a preventative agent for HA-VTE. This case study highlights the importance of

engaging domain expertise while applying machine learning in complex domains.

Index Terms—Venous thromboembolism, machine learning, prediction models, electronic healthcare records, clinical informatics.

I. INTRODUCTION

MACHINE learning (ML) has impacted research in complex domains such as image classification [1], [2], natural language processing [3], [4] and graphs [5]. Increasingly, machine learning has begun to impact domains such as, medicine and healthcare [6], [7]. Machine learning has been deployed to predict coronary artery calcification [8], improve palliative care [9], detect breast ultrasound lesions [10] and drive EHR-based phenotyping [11]. The exponentially growing body of EHR presents an unique opportunity to improve the quality of healthcare services. Machine learning is often conceptualized as a 'black box', capable of learning from data directly without requiring domain expertise, but potentially leading to pitfalls like confounding [12], [13].

We propose a *conversational* approach to machine learning that tightly engages its common practices with domain expertise. By conversational, we do not mean a free-form or natural language conversation, but an exchange of information and methodology between domain experts and machine learning models. This sort of conversation might be realized in many different ways, as we explore in this study. Most machine learning projects focus on engineering a prediction tool which may be less useful in clinical practice where understanding is vital. In medicine, predicting that a patient may be at risk is less actionable than understanding why that patient is at risk. To identify causal relationships in healthcare, machine learning must avoid the pitfalls of confounding variables and surrogate markers (commonly referred together as data-leaking covariates). In some tasks, inferring causality may be much more important than correlation. Prediction only requires correlation, but understanding requires significant knowledge underlying the causal mechanisms. Even in pure prediction problems, interpretability may drive adoption as much as accuracy.

As a case study, we aim to identify incidentally prescribed medicines that influence the risk of hospital-acquired venous thromboembolism (HA-VTE). VTE is a medical condition

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where thrombi (blood clots) form in the venous system due to hypercoagulation. Although thrombi could form in venous vessels of various sizes, deep vein thrombosis (DVT) is the most worrisome. Deep vein thrombi are more likely to break off and follow the circulation to the pulmonary vasculature, causing a life-threatening condition called pulmonary embolism (PE). The incidence of VTE is 1 in 1000 person-years [14] with 10-30% of patients dying in the first month after diagnosis [15]. In the US alone, the disease costs 7 to 10 billion dollars per year [16]. VTE also often recurs, with a community sample estimate of 30% in 10 years [17]. Previous studies on VTE have commonly focused on predicting VTE [18], predicting post-hospitalization [19] and post-operative risk of VTE [20], personalized prediction of VTE risk [21], risk factors that might influence occurrence of VTE [22] and population-scale epidemiology [23]. In the present study, we took a different approach in addressing this critical disease. Rather than developing our own prediction algorithm, we examined the various factors that affect the risk of VTE in hospitalized patients. We focused on identifying interventions, mainly drugs, that reduce the relative risk of VTE by a percentage that is clinically relevant. We found that ondansetron, an anti-nausea medicine, reduces risk of HA-VTE. This finding from healthcare data is consistent with *in vitro* studies showing that ondansetron has anti-aggregation effects on platelets which might suggest a mechanism for its action related to risk reduction in HA-VTE[24]–[27]. Ondansetron is the most widely used drug to treat nausea and vomiting after chemotherapy and surgery [28] with minimal side-effects [29], [30], hence it may be possible to use ondansetron safely to reduce the risk of HA-VTE.

In this research study, we propose an interactive and iterative approach to machine learning that couples domain expertise with model building. One of the critical issues with machine learning in healthcare is the complex underlying relationships between variables in the dataset. Other problems include the shortage of data-points relative to the number of available variables to select from and sparse datasets where many of the labels are missing, posing a significant issue for training supervised learning algorithms. One way to address these shortcomings is to involve human experts in various steps of dataset construction, model training and evaluation [31]. In our work, we iteratively cycle through model training and expert assessment of results to identify and account for confounding variables and surrogate markers (collectively referred to as data-leaking variables) using their clinical reasoning. We also rely on expert knowledge to select positive and negative controls in the evaluation of our model to be confident about the clinical implication of our findings.

This iterative filtering approach is an example of conversational machine learning, in that it brings machine learning into a structured exchange with medical knowledge. We see other ways in which this conversation can be facilitated. In this study, for example, we adopted approaches used in public health to build confidence that the associations found by machine learning are not spurious. The performance of the model in different sub-groups, controlling patient propensity and demonstrating dose-dependency mirror public health analyses. By mirroring

the domain's approach in this way, machine learning leverages domain knowledge in a different way, increasing the robustness of the results in a way that is understandable and interpretable by domain experts.

In this study, we used logistic regression for training models. Logistic regression is an interpretable modeling approach that is widely used in epidemiology. In contrast to epidemiological analyses, where a small number of hand selected features and a tightly defined cohort is used, our logistic regression architecture supports high-parameter modeling. In this work, we used iterative elimination and temporal control to identify data-leaking covariates rather than examining them manually. Finally, the approach proposed in this study can be readily applied to more complex machine learning algorithms.

II. RELATED WORK

In this section, we first provide an overview of the various computational approaches that have been proposed to assess the risk of VTE and develop VTE prophylaxis measures in patient cohorts. Next, we highlight how EHR is playing a crucial role in stimulating quality research in drug repurposing. Last but not least, we provide an overview of interactive and iterative machine learning approaches that are used to improve the interaction between domain experts and machines.

As previously stated, VTE has costly health and economic consequences. Hence, a plethora of research studies had been conducted to understand and predict the risk of VTE in patients. One of the early works in this area studied the post-operative incidence of VTE in patients with leg fractures [20]. In this study, the authors showed that venography is an unreliable outcome measure to assess the clinical burden of symptomatic VTE. Another study by Cushman *et al.* [22] identified the risk factors for VTE. In this study, exogenous factors such as surgery, trauma, immobility, hospitalization, pregnancy and the puerperium (postpartum period after childbirth until the mother's reproductive organs return to a non-pregnant state) and hormone use were found to be associated with significant thrombosis risks. The study also highlighted endogenous VTE risk factors such as cancer, obesity and inherited and acquired hypercoagulation disorders.

Many clinically relevant risk assessment models, built on EHR data using machine learning and statistical analyses, are proposed and made available to help in assessing the risk of VTE in patients. Kawaler *et al.* [19] used naïve Bayes, k-nearest neighbor, support vector machines (SVM) and random forest on EHR to predict which patients are at a higher risk of acquiring post-hospitalization VTE. Law *et al.* [23] used Chi-square tests to carry out a population scale epidemiology among the Han Chinese population in Hong Kong to identify the patients at high risk of VTE. Fresard *et al.* [21] used cost-sensitive classification alongside multi-objective optimization on imbalanced EHR to obtain personalized predictions of VTE risk for high-risk patients. Riordino *et al.* [32] predicted the risk of VTE in the cancer population using candidate bio-markers and risk assessment models. Other studies conducted by Evans *et al.* [33], Spyropoulos *et al.* [34], Hippisley-Cox *et al.* [35] have

proposed their own prediction models to identify the risk of VTE in patients. Qatawneh *et al.* [36] used artificial neural networks to classify the risk of VTE into five risk levels ranging from low to high.

VTE can cause severe complications in hospitalized patients and there is an urgent need for research in VTE and HA-VTE prophylaxis. Previous research studies have shown that, without prophylaxis, HA-VTE is found in 10-40% medical or surgical patients and 74% HA-VTE cases are found in patients with two or more risk factors [37]. Haut *et al.* [38] showed in their study that HA-VTE could largely be prevented through a better understanding of the risk factors that influence HA-VTE and developed protocols for HA-VTE prophylaxis. Mahajerin *et al.* [39] showed increased pediatric HA-VTE cases and identified the risk factors of HA-VTE in children. Other studies [40] showed that hospitals could reduce the risk of HA-VTE by implementing structured HA-VTE prophylaxis with embedded risk assessment and an active surveillance program, called measure-vention, to correct over-prophylaxis and under-prophylaxis. In our study, rather than coming up with our own prediction models to assess the risk of VTE in individual patient, we identified interventions that reduce the risk of HA-VTE in hospitalized patients.

The availability of clinical datasets of large patient cohorts [41] and human genome databases [42] have resulted in undertaking research studies in drug repurposing, that finds new indications for existing drugs. For example, Xu *et al.* [43] showed the use of EHR for validating a recently reported association of metformin as a first-line therapy for type 2 diabetes mellitus (DM2) with reduced cancer mortality. Similarly, we used EHR in our study to identify possible interventions that might reduce the risk of HA-VTE in hospitalized patients.

Interactive machine learning offers a natural way to integrate background knowledge into the modeling stage. Traditional machine learning research has primarily focused on automating feature selection and training. This approach has resulted in significant advances in fields such as in image classification [1] and natural language processing [3]. However, in the healthcare domain, we encounter significant challenges such as small datasets or rare events or complex unknown relationships among variables in the dataset. Studies by Holzinger *et al.* [31], [44], [45] have shown existing challenges with automated machine learning in health informatics and the need to incorporate human and domain knowledge in the learning process. Holzinger *et al.* [31] proposed a change in the concept of *black-box* to *glass-box* machine learning approach, where a human is included in the learning loop for feature selection and labeling the training data. Fails *et al.* [46] proposed the use of tools, like Crayons, to build perceptual user interfaces when using machine learning in interactive settings after automating the feature selection process and using decision trees with sub-sampling techniques. In our study, domain experts are looped in the learning process to make decisions regarding the identification of confounding variables and bio-markers for VTE. Our approach is both interactive and iterative: domain experts interact with machine learning models and identify potential confounding variables and surrogate markers for VTE based on % relative risk reduction (Section III, sub-section C). This interaction is iterated multiple

TABLE I

CHARACTERISTICS OF HOSPITALIZATIONS IN COHORT (QUARTILE-BASED). % VTE POSITIVES ARE BASED ON THE TOTAL VTE POSITIVES IN THE DATASET

Characteristic	Quartile	Median (Max, Min)	%VTE positives
Age(years)	Q4	82.2 (110.4, 74.5)	9.8 (214)
	Q3	68.5 (74.4, 63.2)	13.8 (301)
	Q2	57.7 (63.2, 51)	11.4 (250)
	Q1	39.2 (50.9, 17.9)	8.4 (184)
Length of stay (days)	Q4	8 (214, 5)	90.2 (1969)
	Q3	4 (5, 3)	6 (131)
	Q2	2 (3, 2)	2.8 (63)
	Q1	1 (2, 0)	0.9 (18)
No. of diagnoses	Q4	24 (88,19)	73.8 (1610)
	Q3	16 (19, 13)	17.2 (375)
	Q2	11 (13, 8)	6.9 (150)
	Q1	6 (8, 1)	2.01 (44)
No. of drugs	Q4	22 (101, 17)	79.5 (1731)
	Q3	15 (17, 13)	12.3 (271)
	Q2	11 (13, 9)	5.82 (127)
	Q1	6 (9, 1)	2.38 (48)
No. of procedures	Q4	6 (47, 4)	71.4 (1558)
	Q3	3 (4, 2)	7.2 (158)
	Q2	2 (2, 1)	17.1 (372)
	Q1	1 (1, 1)	0 (0)
Body mass index (BMI)	Q4	38.4 (97.79, 33.46)	26.8 (586)
	Q3	30.3 (33.46, 27.98)	26.1 (569)
	Q2	25.8 (27.98, 23.7)	23.8(519)
	Q1	21.14 (23.7, 3.47)	22 (480)

times until domain experts are confident that almost all the measured confounding variables have been accounted for and surrogate markers are eliminated from the analyses. We validated our findings using proper positive and negative controls throughout our study.

III. MATERIALS & METHODS

We developed logistic regression models that consider diagnoses, drugs, procedures and demographics for each hospitalization as a data-point from our dataset (Table I). Instead of building a model that has high predictive accuracy, we aimed at identifying and eliminating data-leaking variables that mask causal relationships in the data. We propose two approaches: (1) interactive and iterative elimination and accounting for data-leaking covariates via manually examining variables with high RRR values and (2) several control analyses commonly used in public health designed to eliminate spurious associations, such as temporal control via D-dimer lab tests (Section III, sub-section C).

A. Study Population

The data includes records of 397,064 hospitalizations from the BJC healthcare network at St. Louis MO (Table I). The cohort includes 2181 (0.54%) VTE positives and 394,883 (99.45%) VTE negatives. The cohort contains 176,463 (44.44%) male hospitalizations, 189,723 (47.78%) female hospitalizations and 30878 (7.77%) hospitalizations that had no gender information associated with them. Each hospitalization is associated with demographics, diagnoses (23366 ICD 9,10 codes), drugs (1083 unique active ingredients) and procedures (13097 ICD 9-CM, 10-PCS codes). In our study, we included drugs that

were administered orally or intravenous injection or infusion. The cohort contains hospitalizations with a median age of 63.2 years (max: 110.4; min: 17.9). The median length of stay was 3 days (max: 214; min: 0), median number of diagnoses was 13 (max: 88; min: 1), median number of drugs was 13 (max: 101; min: 1), median number of procedures was 2 (max: 47; min: 1) and the median body mass index (BMI) was 27.98 (max: 97.79; min: 3.47). In order to handle outliers in BMI, we excluded height below 50 cm and above 250 cm, and we also excluded weight below 10 kg and above 300 kg, thereby treating BMI of 14713 (3.7%) hospitalizations as missing values. We excluded values greater than 180 days from our analyses and treated them as missing values for the duration of stay. We found 24 out of 397,064 hospitalizations have the length of stay above 180 days and 12 out of 397,064 hospitalizations did not have any length of stay information associated with them. Age was treated as a missing value for the 220 (<0.06 % of the cohort) hospitalizations with age values equal to or greater than 100 years. However, none of these outlier hospitalizations was associated with VTE and was removed from the analyses.

B. Outcome Definition

The VTE index computed by the data provider was used as the definition of positive or negative VTE. Patients were deemed "VTE positive" when there was a (non-primary) diagnosis of VTE made during their inpatient stay. "VTE negative" status was given to those who satisfied the following four criteria: (1) not having VTE diagnosis during their stay; (2) had a primary or secondary VTE diagnosis at presentation; (3) had VTE as primary diagnosis during stay; (4) had major diagnostic category (MDC) as 14 (pregnancy, childbirth and the puerperium). For sensitivity analyses, we studied cancer patients by subdividing our cohort into those with probable cancer diagnoses using SEER 2014 ICD-9-CM Case-finding List [47]. Malignancy results in a 7-fold increase in VTE risk and chemotherapy might have an important effect on other pharmacological modifiers in the risk for VTE.

C. Identification and Elimination of Data-Leaking Covariates

In modeling, the presence of data-leaking covariates (surrogate markers and confounding variables) leads to invalid models [48]. Surrogate markers are variables that are effective at predicting VTE, because the positive outcome directly causes its appearance temporally after VTE diagnosis. Confounding variables are those which encode additional information but overlook other causal relationships that might affect the outcome.

Usually, when there is adequate information on underlying causal relationships between the covariates and the outcome, Pearl's formal system based on causal diagrams is used to account for confounding variables [49], [50]. However, in settings such as in EHR, where we have a large number of interconnected covariates, it is difficult to untangle the underlying causal relationships among the covariates and causal relationships of these covariates with the outcome. Under these circumstances, it is often recommended to control for common causes of exposure

and outcome or to control for pre-exposure measured variables when making decisions regarding controlling for confounding variables. However, in certain settings, controlling for a covariate can introduce bias [51]–[54]. There are several statistical approaches in dealing with confounding variables, such as forward and backward selection. Still, these approaches are unable to distinguish between confounding variables that need to be accounted for and mediating variables that do not need to be accounted for in estimating total effect [55]. A mediating variable links the independent and the dependent variables, providing a mechanism to explain the observed relationship between these variables. In order to handle confounding variables, some degree of domain knowledge is necessary so that an expert may decide which variable can potentially be considered as a confounding variable (and not a mediating variable) or a surrogate marker [56].

In the first approach, in absence of any standardized process to identify or mitigate the effects of these covariates, we opted to review a handful of these covariates rather than manually reviewing all the variables. This approach identified possible leaky covariates based on the risk ratio. Variables with high relative risk reduction (RRR) values were examined and adjudicated based on domain knowledge until no more data-leaking variables were identified. Even though this approach was manual, yet it was efficient as only the variables with high risk ratios were examined.

In the second approach, we deployed a temporal control using D-dimer lab tests. As pointed out by Vanderweele *et al.* [56], statistical analyses, alone, cannot make determinations about temporal order while making decisions about confounding variable selection. Hence, we turned towards domain expertise to look for events in a hospitalization timeline to filter out variables that might potentially be confounding our analyses. Physicians often use D-dimer tests as the first-line assessment of patients suspected of suffering from VTE [57], [58]. D-dimers are protein products of crosslinked fibrin degradation. High levels of D-dimer in the bloodstream is a strong indication of VTE. We identified patients who had undergone D-dimer tests and took the first instance of the D-dimer test as a temporal cut-off. Occurrence of any event (diagnoses, drugs and procedures) after the first D-dimer test was dropped as it might confound our analyses.

D. Modeling and Estimating Relative Risk of Variables

For our study, we used logistic regression, a popular discriminative modeling technique in machine learning. This choice maximizes interpretability to domain experts, because logistic regression is often used to study the relationship between one dependent binary variable and one or more independent variables. Our goal was not to classify patients with the highest accuracy possible, but to identify the associations that are meaningful to domain experts. We chose logistic regression over other popular classification techniques such as decision trees, naïve Bayes, k-nearest neighbors and support vector machines. Aspects of this approach, nonetheless, could be applied to other methods, but that would require further development. For example, variable

importance is estimated by decision forests, which could then be used to direct iterative elimination. However, variable importance is more difficult to extract from SVM models. Our goal was not, ultimately, to advocate for logistic regression over other approaches, but to direct focus in this study towards exploring how a productive exchange with domain expertise could be realized.

One way to improve the interpretability of machine learning algorithms is to translate variable importance into a well-known measure in epidemiology. For example, instead of reporting variable importance as a “weight” as is common in ML, we report it as a relative risk or an odds ratio. While this may seem to be a nuance to epidemiologists, the effort to translate ML terminologies into the language of the medical domain is often neglected in machine learning. In our models, we used odds ratio to approximate the relative risk of a treatment. In epidemiology, relative risk or risk ratio of a treatment is defined as the ratio of probabilities of an event in the treatment group to that in the control group. Relative risk (RR) or risk ratio is calculated as

$$RR = \frac{\frac{a}{a+b}}{\frac{c}{c+d}} \quad (1)$$

where a and b are the respective number of events (VTE positives) and non-events (VTE negatives) in the treatment group and c and d are the respective number of events and non-events in the control group. In our dataset, the number of VTE negatives greatly outweighs the number of VTE positives. Hence, we can rewrite the relative risk as

$$RR = \frac{\frac{a}{a+b}}{\frac{c}{c+d}} \approx \frac{\frac{a}{b}}{\frac{c}{d}} \quad (2)$$

Odds ratio (OR) is defined as the ratio of the odds of an event in the treatment group to the odds of an event in the control group.

$$RR = \frac{\frac{a}{a+b}}{\frac{c}{c+d}} \approx \frac{\frac{a}{b}}{\frac{c}{d}} \approx OR \quad (3)$$

IV. RESULTS

We found ondansetron as a suitable drug for VTE prophylaxis. We used logistic regression for training a series of models guided by domain expertise (Fig. 1). We reported and examined the results from our models and highlighted the key findings regarding relative risk reductions caused by the drugs. We initiated the study by training a model on the entire cohort. The most important variables in the model were identified based on RRR values. Domain experts reviewed these variables and identified those that are likely to be surrogate markers or confounding variables (collectively known as data-leaking covariates). These variables were adjudicated based on domain knowledge. After this step, we re-trained models from scratch and repeated till no more data-leaking variables were identified. This process is not prone to overfitting, because it is not oriented towards increasing classification performance, which decreases each iteration. We specifically studied cancer sub-population and non-cancer sub-population, identified by SEER 2014 ICD-9-CM case-finding

list [47], to ensure that our findings regarding ondansetron are not confounded by cancer sub-population. Based on domain expertise, we used D-dimer tests as a temporal control to filter out medical codes from a hospitalization timeline, as D-dimer lab test is a sensitive indicator of physician’s suspicion of VTE [57]. We carried out cancer sub-population and non-cancer sub-population analyses after filtering out medical codes based on D-dimer lab tests. We have showed that aspirin and ondansetron continue to reduce the risk of HA-VTE across diverse clinical settings, thereby alleviating the concern regarding confounding by cancer sub-populations in the cohort. We further analyzed the dosage response of ondansetron, using aspirin as a positive control and showed that ondansetron has a dose-dependent relative risk reduction on VTE. Finally, we carried out propensity sub-group analysis to account for confounding effects on ondansetron and aspirin by other medical codes, which established confidence in our results and model performance.

A. Iterative Identification of Surrogate Markers and Confounding Variables

Data-leaking covariates mask causal relationships in modeling. An initial model trained with these covariates in the training dataset yielded an area under the receiver operating characteristics curve (AUROC) of 0.92 (10-fold cross-validation AUROC: 0.919), thereby suggesting high predictive accuracy, since these covariates are strongly associated with the outcome variable, i.e. VTE.

We introduced a protocol consisting of several iterations that consisted of analyzing medical codes with extreme RRR by domain experts and epidemiologists to make sure that they were not among the suspected data-leaking covariates. This approach, guided by domain experts and epidemiologists, consisted of an additional pattern-matching step to eliminate medical codes with keywords such as ‘embi’, ‘embolism’, etc. We identified 146 diagnoses (prescribed to 37992 hospitalizations) that included those related to DVT and embolism, 15 drugs (prescribed to 324,962 hospitalizations) that included anti-coagulants and 141 procedures (prescribed to 5972 hospitalizations) identified as data-leaking covariates. Using this approach, we were able to identify diagnosis codes related to VTE as well as blood thinners and anti-coagulants like warfarin and heparin using their RRR percentages (Table II). The percentage of VTE positive hospitalizations (2156 or 98.85%) that were treated using some of these diagnoses, drugs and procedures shows that identifying these variables are critical if we want to capture meaningful relationships in our models. A new model trained on a dataset without data-leaking covariates yielded an AUROC of 0.872 (10-fold cross-validation AUROC: 0.866). The drop in AUROC (Fig. 2A) is expected as the data-leaking variables are either removed or accounted for by the domain experts. If the goal is to uncover causal relationships, then the highest-performing models are not useful, instead models without data leakage are to be preferred. Removal of data-leaking covariates from model input yields substantial changes in causal inferences as is evident from the shift in RRR of drugs such as ondansetron, clopidogrel and pantoprazole (Fig. 2B).

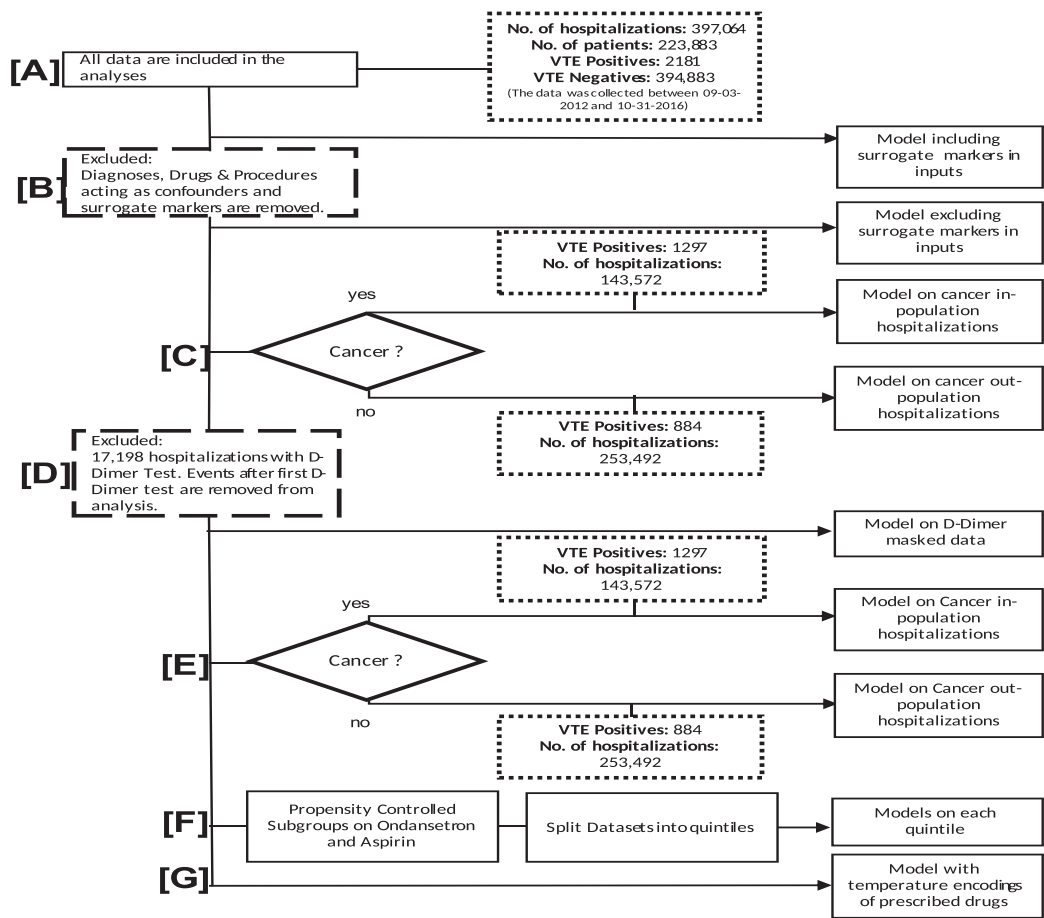


Fig. 1. Detailed development of VTE models using a conversational protocol between machine learning and epidemiological experts. The flow chart shows data-flow, VTE statistics and various domain expertise guided models trained. [A]: At first, all the cohort data were used to train a model. [B]: In the next step, we eliminated the data-leaking covariates to train a subsequent model. [C]: We identified cancer/non-cancer hospitalizations to train two subsequent models on in-cancer and out-cancer sub-populations. [D]: We introduced a temporal control using D-Dimer lab tests as an event in hospitalization timeline and train a model. [E]: Again, we identified cancer in-population and cancer out-population hospitalizations on temporal masked data to train models. [F]: We performed propensity-controlled sub-groups on D-Dimer masked data. [G]: Finally, we conducted dose-dependent analyses on D-Dimer masked data for aspirin and ondansetron.

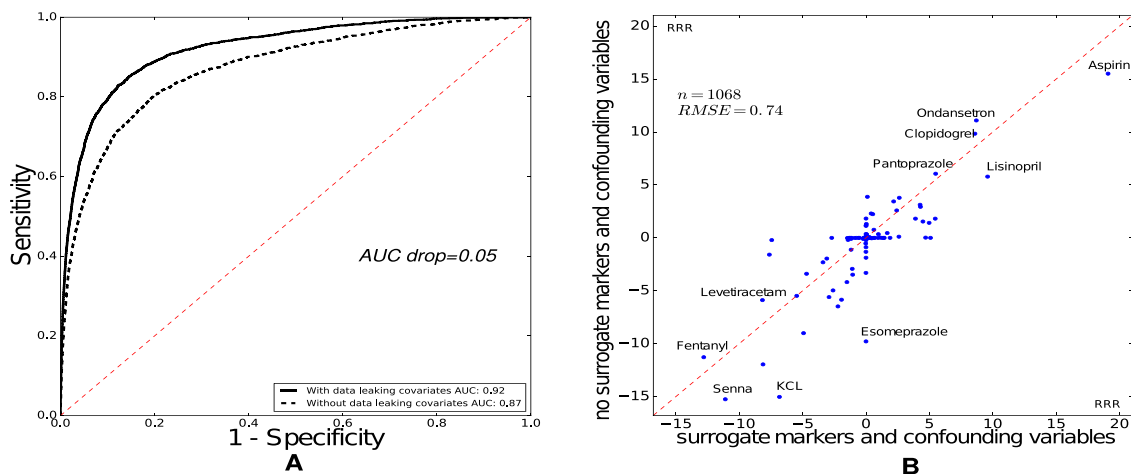


Fig. 2. Variations in model performance and RRR of drugs pre and post elimination of data-leaking covariates. A: AUROC curves of the model predictions before and after data-leaking covariates are eliminated shows a drop in accuracy. B: Changes in % RRR of drugs after data-leaking covariates are eliminated from the dataset show changes in causal inferences.

TABLE II

MEDICAL CODES THAT ARE IDENTIFIED AS DATA-LEAKING COVARIATES. DESCRIPTIONS ARE EXTRACTED FROM THE COHORT DATASET. PERCENTAGE OF VTE POSITIVES ARE BASED ON THE TOTAL VTE POSITIVES IN THE DATASET

data-leaking diagnoses [% VTE positives: 58.7 (1281)]			
ICD code	Description	% RRR	% VTE positives
453.42 (9-CM)	dvt/emb distal lower ext	62.97	34.11 (744)
415.19 (9-CM)	pulm embol/infarct nec	49.72	21.32 (465)
453.41 (9-CM)	dvt/emb prox lower ext	42.73	26.64 (581)
I26.99 (10-CM)	other pulmonary embolism without acute cor pulmonale	13.45	6.83 (149)
415.11 (9-CM)	iatrogen pulm emb/infarc	7.78	4.49 (98)
453.40 (9-CM)	dvt/emblsm lower ext nos	6.65	6.88 (150)
data-leaking drugs [% VTE positives: 97.9 (2136)]			
Drug	Description	RRR	% VTE positives
Warfarin	blood thinner	35.51	32.37 (706)
Heparin	anti-coagulant	25	45.11 (984)
Magnesium sulfate	prevent low blood pressure	23.76	66.62 (1453)
Rivaroxaban	blood thinner	5.23	3.2 (70)
lvp solution potassium	correction of electrolyte/fluid balances	4.54	34.06 (743)
Alteplase	treat acute myocardial infarctions	2.57	17.69 (386)
data-leaking procedures [% VTE positives: 14.3 (312)]			
ICD code	Description	RRR	% VTE positives
38.7 (9-CM)	interruption vena cava	19.41	20.36 (444)
06H03DZ (10-PCS)	insertion of intraluminal device into inferior vena cava, percutaneous approach	1.91	5.23 (114)
88.51 (9-CM)	vena cava angiocardigram	1.63	8.02 (175)
88.41 (9-CM)	contr cerebr arteriogram	0.01	4.22 (92)

TABLE III

VARIATION IN RRR OF DRUGS IN COHORT AND CANCER POPULATION. THESE VARIATIONS HAVE BEEN RECORDED PRE AND POST ELIMINATION OF DATA-LEAKING VARIABLES. VALUES IN PARENTHESES ARE 95% CONFIDENCE INTERVALS. A NEGATIVE VALUE IN THE TABLE INDICATES AN INCREASE IN % RRR FOR THE DRUG

Drug	Cohort population			Cancer population			
	data-leaking variables	w/o data-leaking variables		w/o data-leaking variables			
	w/o temporal control		temporal control	w/o temporal control		temporal control	
				Cancer-in	Cancer-out	Cancer-in	Cancer-out
Aspirin	19.1 (18.1 - 20.1)	15.5 (14.5 - 16.5)	16.2 (15.2 - 17.2)	9.6 (7.7 - 11.4)	22.7 (21.6 - 23.9)	4.3 (2.2 - 6.3)	20 (18.9 - 21.2)
Ondansetron	8.70 (7.2 - 10.1)	11.1 (9.7 - 12.5)	8.9 (7.5 - 10.4)	11.8 (9.6 - 14)	6.9 (5 - 8.8)	8.4 (6.1 - 10.7)	12 (10.3 - 13.8)
Clopidogrel	8.6 (6.1 - 10.9)	9.8 (7.5 - 12.1)	6.8 (4.4 - 9.2)	2.7 (-1.8 - 7)	11.6 (8.8 - 14.3)	1.9 (2.9 - 6.4)	6 (3.2 - 8.9)
Pantoprazole	5.5 (4.1 - 6.8)	6.1 (4.7 - 7.3)	2.2 (0.8 - 3.5)	12 (10.1 - 14)	-0.7 (-2.5 - 1.1)	8.1 (6 - 10.1)	2 (0.3 - 3.6)
Lisinopril	9.6 (7.9 - 11.2)	5.8 (4.1 - 7.3)	7.21 (5.6 - 8.8)	7.5 (4.5 - 10.3)	0.3 (-1.8 - 2.3)	7.2 (4.1 - 10.2)	5.8 (3.9 - 7.7)

B. Drugs With Relative Risk Reduction

Once leaky covariates were eliminated, we were able to identify drugs that significantly modified the risk of VTE by RRR magnitude. Among drugs with the highest RRR are aspirin and ondansetron (RRR of 15.5% and 11%, respectively; Table III). Other drugs also appear to modify the relative risk of VTE such as clopidogrel (RRR 9.8%), pantoprazole (RRR 6.1%) and lisinopril (RRR 5.8%). Ondansetron is identified as a promising candidate for repurposing to VTE prophylaxis. Aspirin, in this analysis, serves as a positive control because it is expected to modify VTE risk [59]–[62]. Moreover, the similarity of ondansetron's RRR to aspirin suggests that the magnitude of the effect is large enough to be clinically useful.

C. D-Dimer Masked Analysis

Removal of data-leaking covariates through iterative refinement is critical to identify causal relationships. In order to filter out events after a possible VTE event, we introduced a temporal control via D-dimer tests. Once the events after D-dimer tests

were eliminated, the model had an AUROC of 0.88 (10-fold cross-validation AUROC: 0.86). Aspirin and ondansetron have a relative risk reduction of around 16.2% and 8.9% respectively on VTE outcome (Table III).

D. Cancer Sub-Population Analysis

We validated that ondansetron, a drug given to cancer patients as an anti-emetic medication, is not confounding our analysis. Using ICD case-finding list [47], cancer patients were identified based on their diagnosis codes in the EHR data. Ondansetron was prescribed to 21.9% (31443) cancer hospitalizations (1297 VTE positives) versus 18.7% (47403) hospitalizations without cancer diagnosis or out-cancer hospitalizations (884 VTE positives). Cancer is usually thought to increase VTE risk. However, it is necessary to confirm that ondansetron continues to reduce the risk of VTE across both in-cancer and out-cancer populations. This sub-population analysis is particularly important, since many hospitalizations have VTE diagnosis at admission and are characterized as VTE negatives. The predictive accuracy

TABLE IV

PHARMACOLOGICAL EFFECTS OF ONDANSETRON AND ASPIRIN WITH VARYING DOSAGES(IN MGS). VALUES IN PARENTHESES ARE 95% CONFIDENCE INTERVALS

Ondansetron				Aspirin		
Dose	Bins(mgs)	visits	RRR(%)	Bins(mgs)	visits	RRR(%)
low	dose≤12	8025	2.32 (0.7 - 4)	dose≤100	72243	4.76 (3.5 - 6)
medium	12<dose≤22	47356	4.93 (1.6 - 8.2)	100<dose≤500	45617	0.64 (-2.62 - 3)
high	22<dose≤ 72	23595	5.47 (-0.9 - 11.4)	dose>500	17360	-4.41 (-11.7 - 2.4)

TABLE V

PROPENSITY CONTROLLED SUB-GROUPS FOR ONDANSETRON AND ASPIRIN. VALUES OF RRR IN PARENTHESES ARE 95% CONFIDENCE INTERVALS. A NEGATIVE VALUE OF RRR IN THE TABLE INDICATES AN INCREASE IN % RRR

Ondansetron propensity controlled sub-groups						
Sub-group	Probability of drug prescription	VTE positives (%)	Visits prescribed(%)	AUC on global D-dimer model	AUC of model trained on sub-group	RRR(%)
1	0.684	27.5(601)	9.5 (37136)	0.87	0.88	21.6 (19.7-23.4)
2	0.538	19.9(436)	3.9 (15410)	0.89	0.89	11.9 (8.6-15)
3	0.462	16.1(351)	2.7 (10801)	0.89	0.88	2.8 (-1.3-6.8)
4	0.394	15.5(340)	2.1 (8102)	0.89	0.89	8.4 (3.9-12.8)
5	0.302	20.3(444)	1.7 (6690)	0.87	0.88	1.1 (-4.3-6.1)
Aspirin propensity controlled sub-groups						
1	0.671	42.7(933)	12.9(50795)	0.89	0.89	29.7 (28.2-31.1)
2	0.563	21.2(464)	8.8(34544)	0.85	0.88	16.2 (14.1-18.1)
3	0.496	16(350)	6.4(24889)	0.87	0.86	14.7 (-0.8-17.1)
4	0.43	11.7(256)	4.6(18081)	0.86	0.85	7 (4.1-9.9)
5	0.334	7.7(169)	2.5(9814)	0.85	0.88	3.1 (-1.3-7.4)

of in-cancer model (AUROC: 0.85) and out-cancer model (AUROC: 0.89) on non-D-dimer masked data were similar to the in-cancer model (AUROC: 0.86) and out-cancer model (AUROC: 0.87) trained on D-dimer masked data. We found that ondansetron continued to have comparable RRR with aspirin both in in-cancer and out-cancer populations (Table III). Cancer is a pro-coagulant state, with increased risk for VTE. The robust reduction and increased incidence in cancer patients suggest that this might be an ideal population to benefit from ondansetron prophylaxis, if this finding is validated.

E. Dosage Dependency

In epidemiological studies, confidence in the modeling results increases if dose dependencies can be demonstrated. Using temperature encoding, our model on ondansetron (AUROC: 0.9, 10-fold cross validation AUROC: 0.89) shows that increase in dosage of ondansetron is associated with a higher RRR on VTE (Table IV). This fits the pattern as increased ondansetron dosages reduce the risk. This suggests that patients at high risk for VTE and who are already on ondansetron, might benefit from higher doses. In contrast, our model (AUROC: 0.9, 10-fold cross validation AUROC: 0.89) shows that increased dosages of aspirin may not always improve pharmacological effects (Table IV). Several studies have shown that aspirin in low doses is more effective in preventing pulmonary embolism than that in higher doses [63], [64]. This pattern is identifiable in this model, serving as a positive control that may increase confidence in the new findings on ondansetron.

F. Propensity Controlled Sub-Groups

After elimination of data-leaking covariates through iterative refinement and temporal cut-off, there is still a concern regarding data-leaking covariates that we might not have already identified.

Propensity score analysis attempts to estimate the effect of a treatment (e.g., ondansetron or aspirin) by accounting for the covariates that help to predict incidence of that treatment [65]. We subdivided the population into five sub-populations by using probabilities from logistic regression models that predict how likely a hospitalized patient would be prescribed either ondansetron or aspirin given with other covariates. We built logistic regression models on each sub-population to predict the likelihood of VTE. Our analyses show that hospitalizations which are more likely to be prescribed ondansetron or aspirin would have high RRR compared to hospitalizations that are less likely to be prescribed either of these two medications. We found that ondansetron's RRR is 9.2 on an average, whereas for aspirin it is around 14.1 (Table V). Again, this enhances the confidence in this key finding.

V. DISCUSSION

In this study, a conversational exchange between machine learning and epidemiology has been made to analyze EHR. The outcome of this study suggests that ondansetron reduces the risk of HA-VTE. The magnitude of ondansetron's RRR is close to that of aspirin, but higher than clopidogrel, a known anti-platelet inhibitor. This risk reduction effect is robust and visible across multiple sub-populations. Previous research studies have shown that Aspirin is the most widely used and tested anti-platelet drug for cardio-vascular diseases [66], [67]. With regard to thromboembolic stroke, the International Stroke Trial [61] and the Chinese Acute Stroke Trial [62], enrolling more than 40,000 patients within 48 hours of the onset of stroke symptoms, suggest that aspirin therapy decreased the risk of recurrent stroke and death without significantly increasing the risk of hemorrhagic stroke. Ondansetron, having similar relative risk reductions on VTE as aspirin, is encouraging. If further validated, these results

might guide improved management of this preventable, yet serious side effect.

This study has primarily focused on EHR, but there is corroborating biochemical evidence that suggests a plausible mechanism of action for this clinical effect. Ondansetron exhibits *in vitro* anti-platelet activity by two distinct signaling pathways: mitogen activated protein-kinase (MAPK) and inositol triphosphate (IP_3) [24]. MAPK inhibition decreases thromboxane A2 production, an important prothrombotic mediator. Inhibition of IP_3 decreases intracellular Ca^{2+} concentration, thereby inhibiting platelet aggregation. Though not established experimentally, it is also possible that ondansetron's antagonism on $5-HT_3$ might interfere with serotonin-mediated platelet activity. Ondansetron has also been shown to decrease post-anesthesia hypotension [68]–[72]. In contrast, increased vasoconstriction raises endothelial dysfunction and is also thrombogenic in action [73]. Therefore, in some contexts, ondansetron may exert a protective effect by reducing vasoconstriction. Several known modifiers of VTE-risk, like aspirin, lisinopril [74] and clopidogrel have also been identified by this analysis, thereby increasing the acceptability of this study. In our study, pantoprazole's RRR is quite high, consistent with a known drug-to-drug interaction that might enhance aspirin's effect [75] and increased blood coagulation time in animal models [76]. This, however, may be the first report establishing a clinical signal associated with RRR for HA-VTE. Use of omeprazole, a drug belonging to the same class, as a negative control, could not be identified by our analysis. This finding is in line with recent studies indicating that omeprazole has no clinical effect, despite the expectation that it might affect VTE risk [77], [78]. If this finding is validated with independent studies, this may indicate the differences in action between pantoprazole and omeprazole, perhaps traceable to the differences in their chemical structure or target profile.

In this study, we are proposing a conversational exchange between machine learning and epidemiology, where insights from both domains can collaborate to build confident causal inferences from EHR. This approach is highly interactive and iterative. Thus, interactive machine learning enables a natural way to couple domain knowledge and expertise into the modeling stage. In the healthcare domain, we often encounter complex unknown underlying relationships between variables in the dataset. Hence, we could not leverage Pearl's formal system [49], [50] based on causal diagrams to decide how to account for confounding variables. Statistical analyses often find it difficult to distinguish between confounding and mediating variables [55]. Under these circumstances, there is a need to apply domain expertise to identify potential data-leaking variables. We propose two approaches: (1) interactive and iterative elimination of data-leaking covariates via manually examining variables with high RRR values and (2) temporal control via D-dimer lab tests. These approaches are instrumental in this scenario, where we are leveraging domain expertise in deploying machine learning solutions to critical healthcare problems. Application of automatic machine learning methods in complex healthcare domain remains elusive in the near future. Gaussian processes with standard kernel machines struggle on function extrapolation problems that are quite trivial for humans. Wilson

et al. [79] proposed a promising human-in-the-loop approach via human kernels which were able to utilize human cognitive abilities and perform much better than standard kernels. We believe that incorporating domain expertise alongside machine learning in an interactive setting can be of particular interest when solving critical problems in healthcare.

Machine learning is a powerful technique, but it is limited by the same pitfalls as epidemiological research and the same mitigation strategies can be applied to overcome these pitfalls. As with any clinical study, confounding factors impact the analyses, creating both false negatives and false positives. Interactive and iterative identification of data-leaking covariates and temporal control via D-dimer lab tests engage domain expertise in an exchange with machine learning, proving to be an effective way of handling these variables. Still, variables weakly related to the outcome but strongly related to a medication might still produce meaningful bias without being identified by our approach. Also, our approach ignores variation of these variables. In the case of VTE, diagnoses and procedures are expected to be the strongest confounding variables and would also be captured well in an EHR dataset. Ensuring robustness of results in epidemiologically motivated sub-groups increases confidence in our results. Of course, for questions in which other pieces of information are vital, such as environmental or behavioral variables, accurate inferences are unlikely to be made based solely on data from an EHR. Furthermore, logistic regression suffers from outliers and multicollinearity. Multicollinearity can lead to unstable estimates and inaccurate variances which, in turn, affect confidence intervals and hypothesis tests. Machine learning is not a panacea, but it might take into account more information about the contextual details of each patient.

VI. CONCLUSION

This case study is an example of conversation between machine learning and domain expertise in healthcare. Here, it uncovered new knowledge about ondansetron, which might improve clinical management of a severe complication in hospitalized patients. This finding is robust to several control analyses. A prospective study of hospitalized patients would clarify its clinical utility.

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N.L.D identified the surrogate markers. K.F.W suggested the project and provided the data. S.J.S oversaw all aspects of the project.

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